



St. MARTIN'S ENGINEERING COLLEGE

UGC Autonomous
NBA & NAAC A+ ACCREDITED
Dhulapally, Secunderabad-500100.



Department of Information Technology

Year: IV

Semester: I

Section: C

Mini Project Title and Abstract

**NEURAL NETWORK-BASED PREDICTIVE MODELING FOR
COMPLEX PSYCHIATRIC DISORDER IDENTIFICATION USING
EEG FEATURES**

ABSTRACT

This project presents a deep learning approach using convolutional neural networks (CNNs) to identify complex psychiatric disorders such as depression, bipolar disorder, and schizophrenia through EEG (electroencephalogram) signals. EEG data, which captures electrical activity in the brain, is first cleaned and transformed into visual formats like spectrograms and topographic maps to highlight patterns across different brain regions and frequency bands. These images are then fed into a CNN model that automatically learns to detect subtle differences in brainwave activity between healthy individuals and those with psychiatric conditions. The model is trained on labeled EEG datasets and optimized using techniques like dropout and batch normalization to improve accuracy and prevent overfitting. Results show that the CNN can reliably classify disorders with high precision, identifying key biomarkers in theta, alpha, and beta bands. This method offers a non-invasive, scalable, and interpretable solution for early diagnosis and clinical decision-making, potentially improving treatment outcomes and reducing diagnostic delays in mental health care.

BATCH NO:17

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